EXPERIMENT - 4

AIM

Using the Employee schema, write and execute SQL queries with INNER JOIN, LEFT JOIN, self-join, 3-way join, correlated subqueries, EXISTS, and simulated INTERSECT and EXCEPT. Compare the execution plans of selected queries using EXPLAIN.

OBJECTIVES

• To understand how different types of JOIN operations combine rows from related tables.

• To perform a self-join on the Employee table to compare employees within the same department.

• To use correlated subqueries and EXISTS for row-by-row and existence-based filtering.

• To simulate INTERSECT and EXCEPT operations using subqueries supported by MySQL.

• To inspect and compare query execution plans using EXPLAIN.

DATABASE USED

The queries use the CompanyDB database and the Employee–Department–Project schema from the reference experiment. The reference document defines 30 employees, 5 departments and 8 projects. The Employee table contains emp_id, emp_name, salary, hire_date, dept_id and project_id, with foreign keys to Department and Project.

```
USE CompanyDB;
```

SCHEMA REFERENCE

Table	Important Columns	Relationship
Employee	emp_id, emp_name, salary, hire_date, dept_id, project_id	dept_id → Department; project_id → Project
Department	dept_id, dept_name	Primary key: dept_id
Project	project_id, project_name, budget, dept_id	Primary key: project_id; dept_id → Department

The source document creates Department, Project and Employee in this order and establishes the foreign-key relationships shown above.

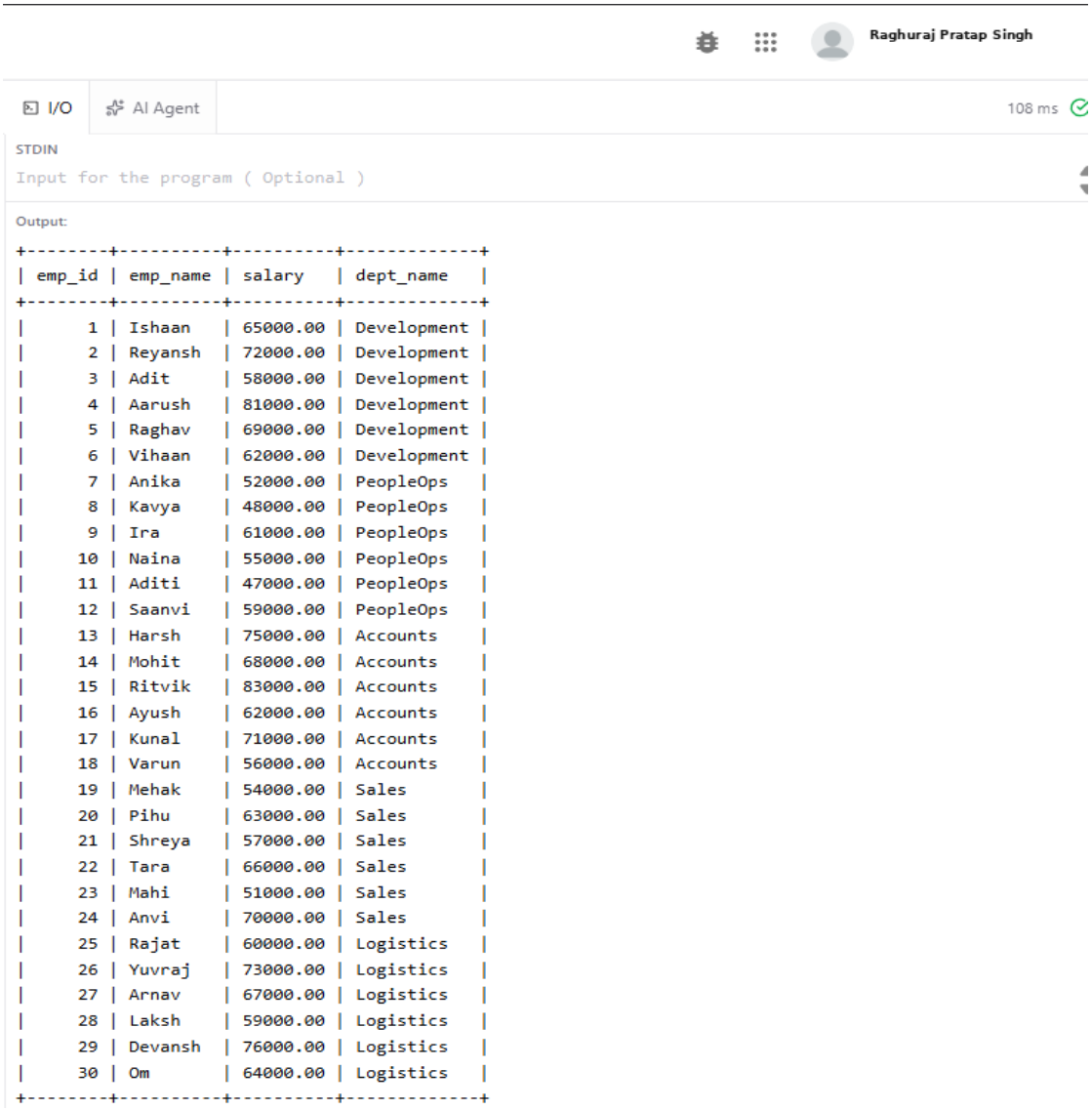
SOURCE CODE / SQL QUERIES

1. INNER JOIN

Display employee names, salaries and their department names using an INNER JOIN.

```
SELECT
    e.emp_id,
    e.emp_name,
    e.salary,
    d.dept_name
FROM Employee e
INNER JOIN Department d
    ON e.dept_id = d.dept_id
ORDER BY e.emp_id;
```

Output




```
+--------+----------+----------+-------------+
| emp_id | emp_name | salary   | dept_name   |
+--------+----------+----------+-------------+
|      1 | Ishaan   | 65000.00 | Development |
|      2 | Reyansh  | 72000.00 | Development |
|      3 | Adit     | 58000.00 | Development |
|      4 | Aarush   | 81000.00 | Development |
|      5 | Raghav   | 69000.00 | Development |
|      6 | Vihaan   | 62000.00 | Development |
|      7 | Anika    | 52000.00 | PeopleOps   |
|      8 | Kavya    | 48000.00 | PeopleOps   |
|      9 | Ira      | 61000.00 | PeopleOps   |
|     10 | Naina    | 55000.00 | PeopleOps   |
|     11 | Aditi    | 47000.00 | PeopleOps   |
|     12 | Saanvi   | 59000.00 | PeopleOps   |
|     13 | Harsh    | 75000.00 | Accounts    |
|     14 | Mohit    | 68000.00 | Accounts    |
|     15 | Ritvik   | 83000.00 | Accounts    |
|     16 | Ayush    | 62000.00 | Accounts    |
|     17 | Kunal    | 71000.00 | Accounts    |
|     18 | Varun    | 56000.00 | Accounts    |
|     19 | Mehak    | 54000.00 | Sales       |
|     20 | Pihu     | 63000.00 | Sales       |
|     21 | Shreya   | 57000.00 | Sales       |
|     22 | Tara     | 66000.00 | Sales       |
|     23 | Mahi     | 51000.00 | Sales       |
|     24 | Anvi     | 70000.00 | Sales       |
|     25 | Rajat    | 60000.00 | Logistics   |
|     26 | Yuvraj   | 73000.00 | Logistics   |
|     27 | Arnav    | 67000.00 | Logistics   |
|     28 | Laksh    | 59000.00 | Logistics   |
|     29 | Devansh  | 76000.00 | Logistics   |
|     30 | Om       | 64000.00 | Logistics   |
+--------+----------+----------+-------------+
```

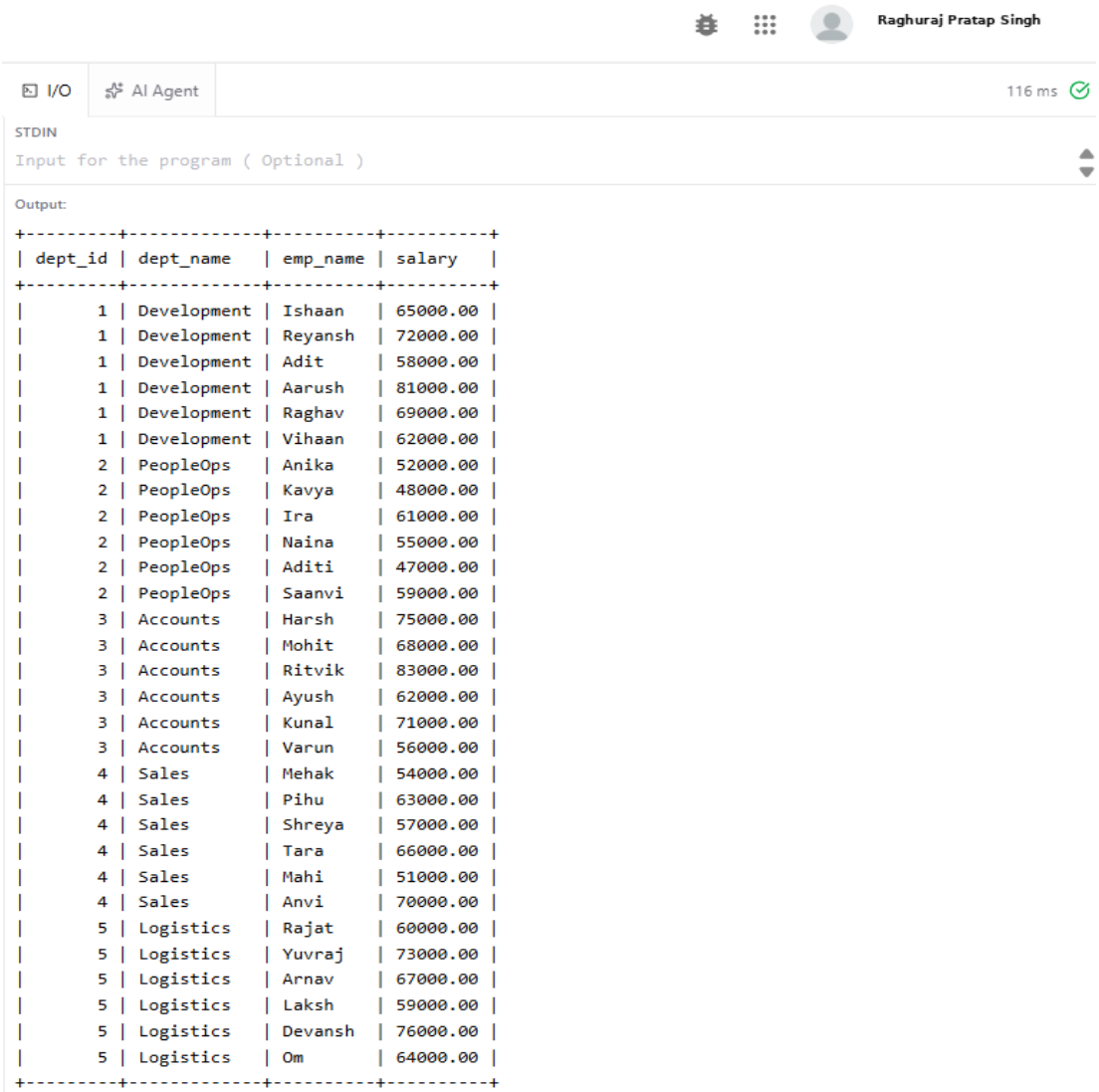
The full query returns all 30 employees. The Department values come from the Department table in the reference data.

2. LEFT JOIN

Display all departments and the employees belonging to them. LEFT JOIN keeps every department from the left table even if no matching employee exists.

```
SELECT
    d.dept_id,
    d.dept_name,
    e.emp_name,
    e.salary
FROM Department d
LEFT JOIN Employee e
    ON d.dept_id = e.dept_id
ORDER BY d.dept_id, e.emp_id;
```

OUTPUT



dept_id	dept_name	emp_name	salary
1	Development	Ishaan	65000.00
1	Development	Reyansh	72000.00
1	Development	Adit	58000.00
1	Development	Aarush	81000.00
1	Development	Raghav	69000.00
1	Development	Vihaan	62000.00
2	PeopleOps	Anika	52000.00
2	PeopleOps	Kavya	48000.00
2	PeopleOps	Ira	61000.00
2	PeopleOps	Naina	55000.00
2	PeopleOps	Aditi	47000.00
2	PeopleOps	Saanvi	59000.00
3	Accounts	Harsh	75000.00
3	Accounts	Mohit	68000.00
3	Accounts	Ritvik	83000.00
3	Accounts	Ayush	62000.00
3	Accounts	Kunal	71000.00
3	Accounts	Varun	56000.00
4	Sales	Mehak	54000.00
4	Sales	Pihu	63000.00
4	Sales	Shreya	57000.00
4	Sales	Tara	66000.00
4	Sales	Mahi	51000.00
4	Sales	Anvi	70000.00
5	Logistics	Rajat	60000.00
5	Logistics	Yuvraj	73000.00
5	Logistics	Arnav	67000.00
5	Logistics	Laksh	59000.00
5	Logistics	Devansh	76000.00
5	Logistics	Om	64000.00

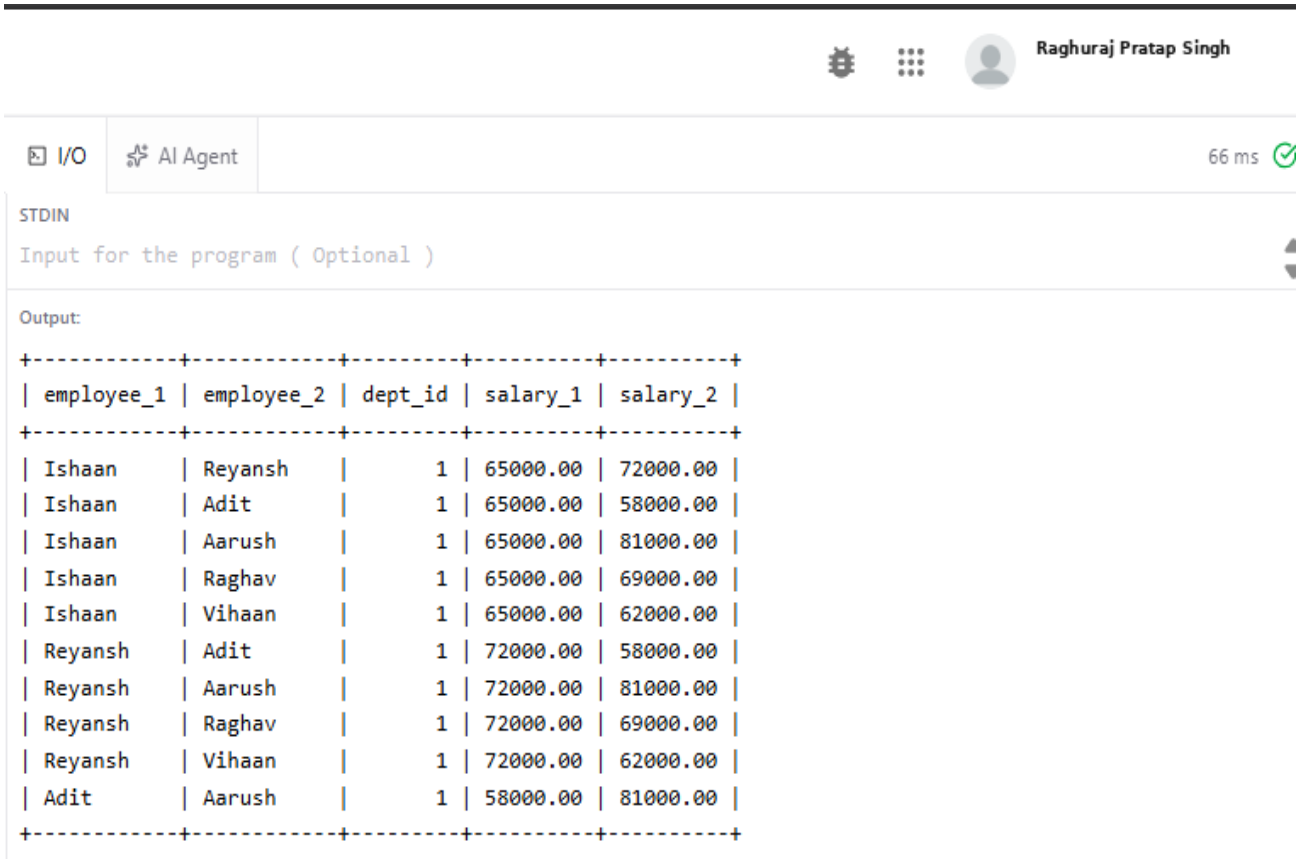
In the supplied data, all five departments have employees, so no NULL employee row appears. The LEFT JOIN would show a department with NULL employee columns if such a department existed.

3. SELF-JOIN

Compare two employees belonging to the same department. The Employee table is joined to itself using two aliases.

```
SELECT
    e1.emp_name AS employee_1,
    e2.emp_name AS employee_2,
    e1.dept_id,
    e1.salary AS salary_1,
    e2.salary AS salary_2
FROM Employee e1
INNER JOIN Employee e2
    ON e1.dept_id = e2.dept_id
   AND e1.emp_id < e2.emp_id
ORDER BY e1.dept_id, e1.emp_id, e2.emp_id
LIMIT 10;
```

OUTPUT (first 10 rows)



employee_1	employee_2	dept_id	salary_1	salary_2
Ishaan	Reyansh	1	65000.00	72000.00
Ishaan	Adit	1	65000.00	58000.00
Ishaan	Aarush	1	65000.00	81000.00
Ishaan	Raghav	1	65000.00	69000.00
Ishaan	Vihaan	1	65000.00	62000.00
Reyansh	Adit	1	72000.00	58000.00
Reyansh	Aarush	1	72000.00	81000.00
Reyansh	Raghav	1	72000.00	69000.00
Reyansh	Vihaan	1	72000.00	62000.00
Adit	Aarush	1	58000.00	81000.00

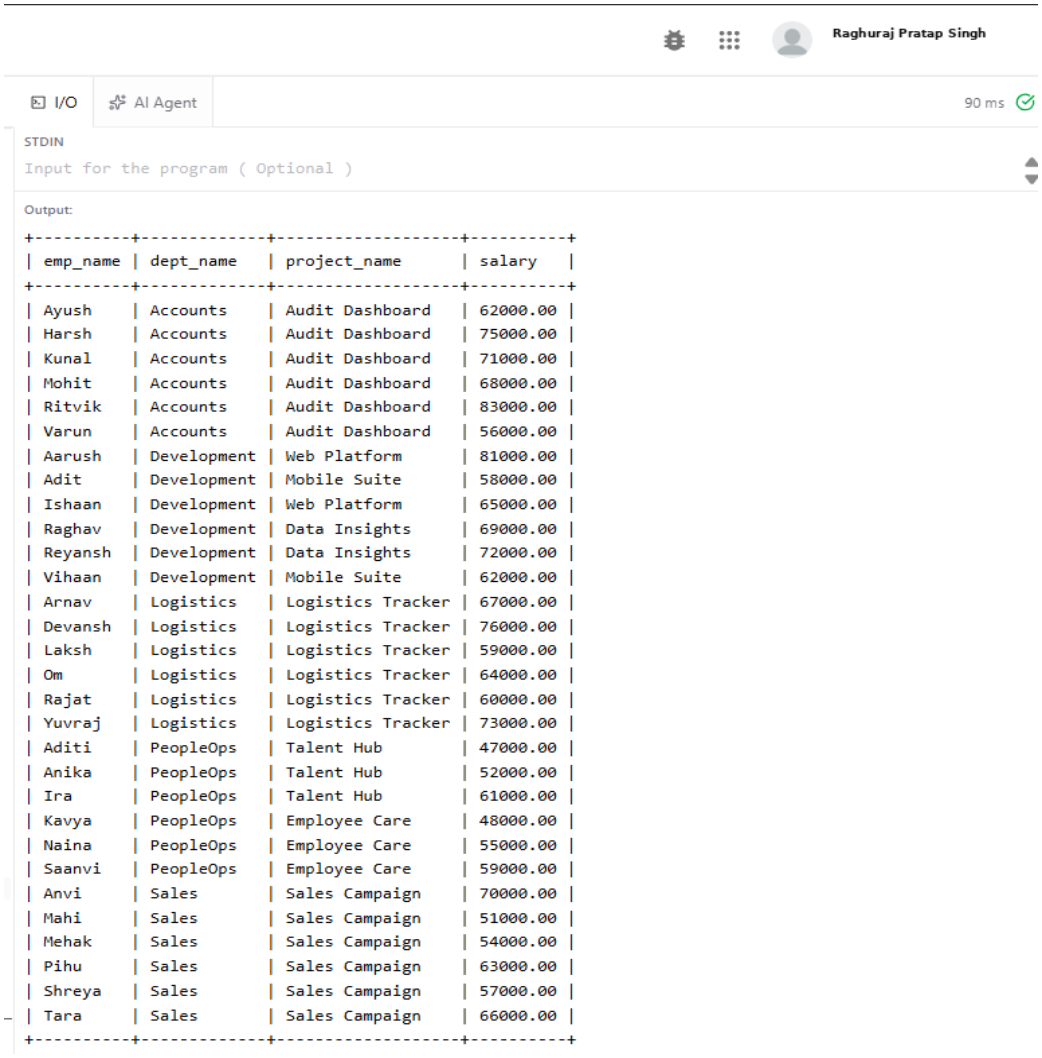
The condition e1.emp_id < e2.emp_id prevents the same pair from being repeated in reverse order and prevents an employee from being paired with itself.

4. 3-WAY JOIN

Display employee name, department, project and salary by joining Employee, Department and Project.

```
SELECT
    e.emp_name,
    d.dept_name,
    p.project_name,
    e.salary
FROM Employee e
INNER JOIN Department d
    ON e.dept_id = d.dept_id
INNER JOIN Project p
    ON e.project_id = p.project_id
ORDER BY d.dept_name, e.emp_name;
```

OUTPUT



emp_name	dept_name	project_name	salary
Ayush	Accounts	Audit Dashboard	62000.00
Harsh	Accounts	Audit Dashboard	75000.00
Kunal	Accounts	Audit Dashboard	71000.00
Mohit	Accounts	Audit Dashboard	68000.00
Ritvik	Accounts	Audit Dashboard	83000.00
Varun	Accounts	Audit Dashboard	56000.00
Aarush	Development	Web Platform	81000.00
Adit	Development	Mobile Suite	58000.00
Ishaan	Development	Web Platform	65000.00
Raghav	Development	Data Insights	69000.00
Reyansh	Development	Data Insights	72000.00
Vihaan	Development	Mobile Suite	62000.00
Arnav	Logistics	Logistics Tracker	67000.00
Devansh	Logistics	Logistics Tracker	76000.00
Laksh	Logistics	Logistics Tracker	59000.00
Om	Logistics	Logistics Tracker	64000.00
Rajat	Logistics	Logistics Tracker	60000.00
Yuvraj	Logistics	Logistics Tracker	73000.00
Aditi	PeopleOps	Talent Hub	47000.00
Anika	PeopleOps	Talent Hub	52000.00
Ira	PeopleOps	Talent Hub	61000.00
Kavya	PeopleOps	Employee Care	48000.00
Naina	PeopleOps	Employee Care	55000.00
Saanvi	PeopleOps	Employee Care	59000.00
Anvi	Sales	Sales Campaign	70000.00
Mahi	Sales	Sales Campaign	51000.00
Mehak	Sales	Sales Campaign	54000.00
Pihu	Sales	Sales Campaign	63000.00
Shreya	Sales	Sales Campaign	57000.00
Tara	Sales	Sales Campaign	66000.00

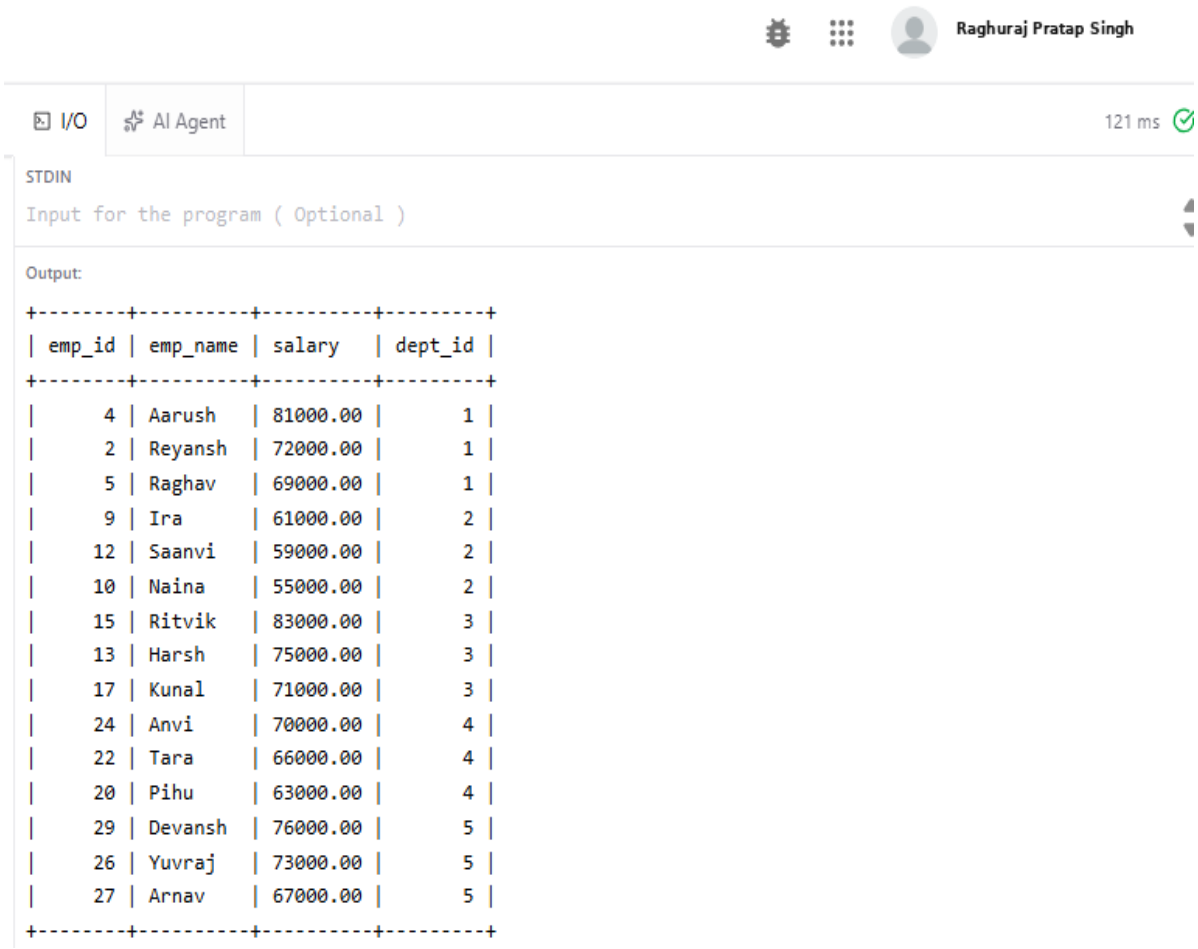
The reference experiment uses the same three-table join pattern and orders the result by department and employee name.

5. CORRELATED SUBQUERY

Display employees whose salary is greater than the average salary of their own department. The inner query refers to the outer Employee row, making it a correlated subquery.

```
SELECT
   e.emp_id,
   e.emp_name,
   e.salary,
   e.dept_id
FROM Employee e
WHERE e.salary > (
   SELECT AVG(e2.salary)
   FROM Employee e2
   WHERE e2.dept_id = e.dept_id
)
ORDER BY e.dept_id, e.salary DESC;
```

OUTPUT



emp_id	emp_name	salary	dept_id
4	Aarush	81000.00	1
2	Reyansh	72000.00	1
5	Raghav	69000.00	1
9	Ira	61000.00	2
12	Saanvi	59000.00	2
10	Naina	55000.00	2
15	Ritvik	83000.00	3
13	Harsh	75000.00	3
17	Kunal	71000.00	3
24	Anvi	70000.00	4
22	Tara	66000.00	4
20	Pihu	63000.00	4
29	Devansh	76000.00	5
26	Yuvraj	73000.00	5
27	Arnav	67000.00	5

The department averages used by this query are consistent with the reference experiment's GROUP BY/HAVING results, including IT 67833.33, Finance 69166.67 and Operations 66500.00.

6. EXISTS

Find departments that have at least one employee with a salary greater than 80,000.

```
SELECT
   d.dept_id,
   d.dept_name
FROM Department d
WHERE EXISTS (
   SELECT 1
   FROM Employee e
   WHERE e.dept_id = d.dept_id
    AND e.salary > 80000
)
ORDER BY d.dept_id;
```

OUTPUT

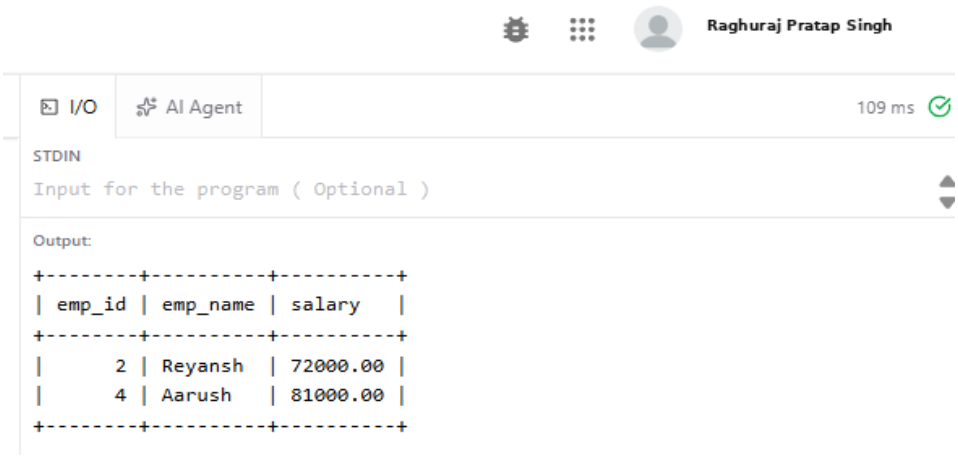


EXISTS returns TRUE as soon as the subquery finds at least one qualifying employee for the current department.

7. SIMULATED INTERSECT

MySQL can express the intersection of two employee sets by requiring the employee ID to occur in both subqueries. Here, the two sets are: employees in IT and employees whose salary is greater than 70,000.

```
SELECT emp_id, emp_name, salary
FROM Employee
WHERE emp_id IN (
   SELECT emp_id
   FROM Employee
   WHERE dept_id = 1
)
AND emp_id IN (
   SELECT emp_id
   FROM Employee
   WHERE salary > 70000
)
ORDER BY emp_id;
```

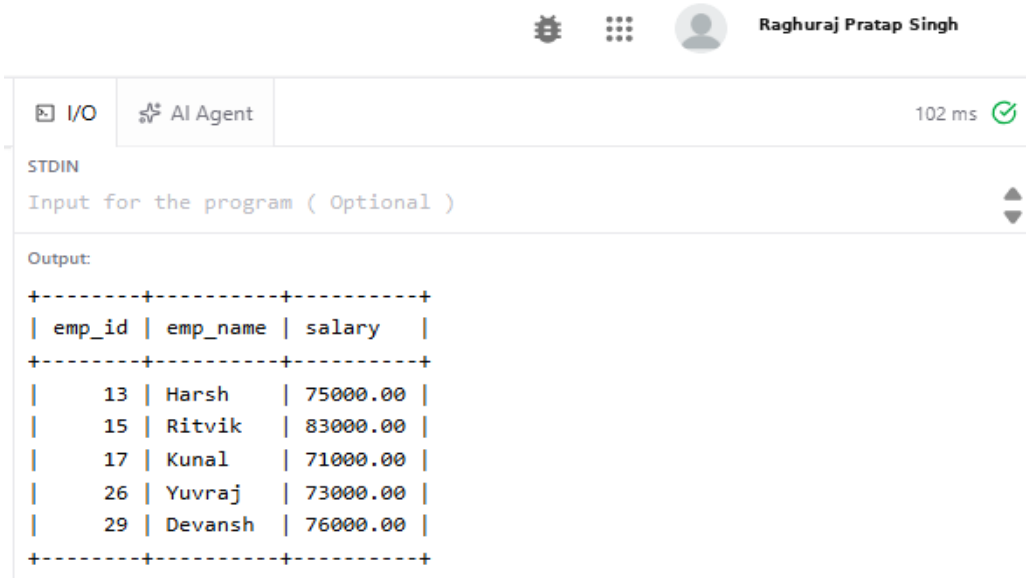


The result contains only employees that satisfy both sets, which is the behavior of an INTERSECT operation.

8. SIMULATED EXCEPT

Simulate EXCEPT by selecting employees with salary greater than 70,000 while excluding employees belonging to IT.

```
SELECT emp_id, emp_name, salary
FROM Employee
WHERE emp_id IN (
   SELECT emp_id
   FROM Employee
   WHERE salary > 70000
)
AND emp_id NOT IN (
   SELECT emp_id
   FROM Employee
   WHERE dept_id = 1
)
ORDER BY emp_id;
```



9. EXPLAIN – EXECUTION PLAN

EXPLAIN shows how the MySQL optimizer intends to execute a SELECT statement. It can be used to inspect access type, possible indexes, selected index, estimated rows and additional operations.

A. EXPLAIN for INNER JOIN

```
EXPLAIN
SELECT
   e.emp_name,
   d.dept_name
FROM Employee e
INNER JOIN Department d
   ON e.dept_id = d.dept_id
WHERE e.salary > 70000;
```

B. EXPLAIN for Correlated Subquery

```
EXPLAIN
SELECT
   e.emp_name,
   e.salary
FROM Employee e
WHERE e.salary > (
   SELECT AVG(e2.salary)
   FROM Employee e2
   WHERE e2.dept_id = e.dept_id
);
```

C. EXPLAIN for EXISTS

```
EXPLAIN
SELECT d.dept_name
FROM Department d
WHERE EXISTS (
   SELECT 1
   FROM Employee e
   WHERE e.dept_id = d.dept_id
     AND e.salary > 80000
);
```

10. COMPARISON OF EXECUTION PLANS

Query	Main operation	Plan aspect to compare	Expected observation
INNER JOIN	Employee ↔ Department	Join access on dept_id	Checks how Employee and Department are accessed and joined.
Correlated subquery	Outer Employee + repeated department AVG logic	Subquery execution and rows examined	May require more work because the inner query is logically related to each outer row.
EXISTS	Department + existence test on Employee	Early existence check and filtering	Can stop searching once a qualifying Employee row is found.

For a small dataset such as the 30 employees in this experiment, the optimizer may choose table scans because the cost is low. With larger tables and suitable indexes on join/filter columns, the execution plan can change. The comparison should therefore be based on the actual EXPLAIN output produced in the lab environment.

RESULT

The Employee–Department–Project database was used to successfully demonstrate INNER JOIN, LEFT JOIN, self-join, 3-way JOIN, correlated subqueries, EXISTS, simulated INTERSECT and simulated EXCEPT. EXPLAIN statements were also prepared to compare how MySQL plans the different query forms.

CONCLUSION

Different SQL constructs can produce related results while using different execution strategies. JOIN operations combine rows from tables, correlated subqueries evaluate related values for outer rows, EXISTS tests whether matching rows exist, and subqueries can simulate set operations such as INTERSECT and EXCEPT. EXPLAIN provides information needed to study and compare these execution strategies.